Multiple estimates

Model. Let $\mathcal{D}_n = \{\mathbf{s}_1, \dots, \mathbf{s}_n\}$ be the set of observed locations, $n_{\mathbf{s}}$ the number of estimates at location $\mathbf{s} \in \mathcal{D}_n$, and $N = \sum_{\mathbf{s}\in\mathcal{D}_n} n_{\mathbf{s}}$. Let $Y(\mathbf{s})_j$ denote the jth estimate of the logit proportion at $\mathbf{s}$, with covariate vector $X(\mathbf{s})_j \in \mathbb{R}^p$. We assume

$$Y(\mathbf{s})_j = X(\mathbf{s})_j^\top \beta + Z(\mathbf{s}) + \epsilon(\mathbf{s})_j, \qquad \epsilon(\mathbf{s})_j \overset{iid}{\sim} N(0, \sigma_\epsilon^2) \;\perp\; Z, \qquad \mathbf{s} \in \mathcal{D}_n,\ j = 1, \dots, n_{\mathbf{s}}, \tag{1}$$

where Z is a zero-mean Gaussian stationary isotropic process with $c_Z \in \mathcal{D}_\infty$. Hence

$$C_Y\big((\mathbf{s}, j), (\mathbf{s}', j')\big) := \mathrm{Cov}\big(Y(\mathbf{s})_j, Y(\mathbf{s}')_{j'}\big) = c_Z\big(\|\mathbf{s} - \mathbf{s}'\|_2\big) + \sigma_\epsilon^2\, 1\{\mathbf{s} = \mathbf{s}',\, j = j'\}. \tag{2}$$

Mixture representation. Since $c_Z \in \mathcal{D}_\infty$,

$$c_Z(h) = \int_{[0,\infty)} K_G(\alpha, h)\, d\mu_Z(\alpha), \tag{3}$$

where $K_G(0, h) = 1\{h = 0\}$, $K_G(\alpha, h) = \exp(-h^2/\alpha^2)$ for $\alpha > 0$, and μ_Z is a positive measure on $[0, \infty)$.

$$C_Y\big((\mathbf{s}, j), (\mathbf{s}', j')\big) = \int_{[0,\infty)} K_G\big(\alpha, \|\mathbf{s} - \mathbf{s}'\|_2\big)\, d\mu_Z(\alpha) + \sigma_\epsilon^2\, 1\{\mathbf{s} = \mathbf{s}',\, j = j'\}. \tag{4}$$

Note that $\mu_Z(\{0\})$ enters through $K_G(0, \|\mathbf{s}-\mathbf{s}'\|_2) = 1\{\mathbf{s} = \mathbf{s}'\}$ and is a location effect shared by all estimates at $\mathbf{s}$, whereas σ_ϵ^2 is measurement error specific to each estimate. They both correspond to $h = 0$. With one estimate per location the two kernels coincide, only $\mu_Z(\{0\}) + \sigma_\epsilon^2$ is identifiable, and (4) reduces to equation (5) of the main paper with $\mu_Y = \mu_Z + \sigma_\epsilon^2 \delta_0$. With repeated estimates they are separated by the within-location covariance $\mathrm{Cov}(Y(\mathbf{s})_j, Y(\mathbf{s})_{j'}) = c_Z(0)$, $j \neq j'$, as opposed to $\mathrm{Var}(Y(\mathbf{s})_j) = c_Z(0) + \sigma_\epsilon^2$. We therefore keep μ_Z and σ_ϵ^2 separate and estimate both.

Finite-grid approximation. Let $\mathbf{Y} \in \mathbb{R}^N$ and $X \in \mathbb{R}^{N\times p}$ be the logit proportion estimates and their covariates. Let $R \in \{0,1\}^{N\times n}$ be the incidence matrix with $R_{(\mathbf{s},j),k} = 1\{\mathbf{s} = \mathbf{s}_k\}$ and let $D \in \mathbb{R}^{n\times n}$ be the distance matrix of the unique locations. Let s_h be the maximum pairwise distance among the locations in $\mathcal{D}_n$, $\tilde{h}_{\mathbf{s}\mathbf{s}'} = \|\mathbf{s} - \mathbf{s}'\|_2 / s_h$, $\tilde{D} = D/s_h$, $\tilde{\alpha} = \alpha/s_h$, and $\tilde{\mu}_Z$ the measure induced from μ_Z by this change of variable. Given a grid $\{\tilde{\alpha}_1, \dots, \tilde{\alpha}_s\}$ with $\tilde{\alpha}_1 = 0$, with n the number of unique locations, we approximate $\tilde{\mu}_Z$ by

$$\tilde{\mu}_Z(\cdot; \theta) := \sum_{i=1}^{s} \theta_i\, \delta_{\{\tilde{\alpha}_i\}}, \qquad \theta_1, \dots, \theta_s \geq 0, \tag{5}$$

where $\theta_1 = \tilde{\mu}_Z(\{0\}; \theta)$ is the location-effect variance. We write $\theta_0 := \sigma_\epsilon^2$ and $\theta := (\theta_0, \theta_1, \dots, \theta_s)^\top \in \mathbb{R}^{s+1}_{\geq 0}$. This gives

$$C_Y\big((\mathbf{s}, j), (\mathbf{s}', j'); \theta\big) = \theta_0\, 1\{\mathbf{s} = \mathbf{s}',\, j = j'\} + \theta_1\, 1\{\mathbf{s} = \mathbf{s}'\} + \sum_{i=2}^{s} \theta_i \exp\big(-\tilde{h}^2_{\mathbf{s}\mathbf{s}'}/\tilde{\alpha}_i^2\big), \tag{6}$$

and, in matrix form, with $\Sigma_G(\tilde{\alpha}_i) \in \mathbb{R}^{n\times n}$ as in equation (13) of the main paper evaluated on $\tilde{D}$, so that $\Sigma_G(\tilde{\alpha}_1) = I_n$,

$$\Sigma(\theta) := \sum_{i=0}^{s} \theta_i \Sigma_i, \qquad \Sigma_0 := I_N, \qquad \Sigma_i := R\, \Sigma_G(\tilde{\alpha}_i)\, R^\top, \quad i = 1, \dots, s, \tag{7}$$

where $\Sigma_1 = RR^\top$ is block diagonal with blocks $J_{n_\mathbf{s}}$, the $n_\mathbf{s} \times n_\mathbf{s}$ matrix of ones.

Note: When $n_\mathbf{s} = 1$ for all $\mathbf{s}$, $R = I_n$ and $\Sigma_0 = \Sigma_1 = I_n$, so only $\theta_0 + \theta_1$ is identifiable and (7) reduces to equation (12) of the main paper. With repeated estimates $\Sigma_0 \neq \Sigma_1$ and both are estimated.

Prediction. Let $\hat{\theta} = (\hat{\theta}_0, \hat{\theta}_1, \ldots, \hat{\theta}_s)^\top$ and $\hat{\beta} = \hat{\beta}_{\Sigma(\hat{\theta})}$ be the estimates obtained from the data, and define the fitted covariance function and the corresponding matrices

$$\hat{c}_Z(\tilde{h}) = \hat{\theta}_1 \, 1\{\tilde{h} = 0\} + \sum_{i=2}^{s} \hat{\theta}_i \exp\big(-\tilde{h}^2/(\tilde{\alpha}_i)^2\big), \qquad \hat{\Sigma}_G := \hat{c}_Z(D) = \hat{\theta}_1 I_n + \sum_{i=2}^{s} \hat{\theta}_i \Sigma_G(\tilde{\alpha}_i), \tag{8}$$

so that $\Sigma(\hat{\theta}) = \hat{\theta}_0 I_N + R\hat{\Sigma}_G R^\top$. The measurement-error weight $\hat{\theta}_0$ enters only through the diagonal of $\Sigma(\hat{\theta})$, since ϵ is independent of Z and across estimates.

For a sub-national area $\mathbf{s}_0$, observed or not, $X_{\text{ref}}(\mathbf{s}_0)$ holds the reference covariate values used for the map with final estimates (in cross-validation, the held-out estimate's own covariates). Measurement error has mean zero and is independent of everything else, so it does not enter the point prediction. Let $r(\mathbf{s})_j = Y(\mathbf{s})_j - X(\mathbf{s})_j^\top \hat{\beta}$ be the residuals formed with each estimate's own covariates (observation/location level). The kriging predictor (19) of the main paper becomes

$$\begin{aligned} \hat{Y}(\mathbf{s}_0) &= X_{\text{ref}}(\mathbf{s}_0)^\top \hat{\beta} + \hat{c}_0^\top \Sigma(\hat{\theta})^{-1} \mathbf{r}, \qquad \hat{c}_0 := \operatorname{Cov}\big(Z(\mathbf{s}_0), \mathbf{Y}\big) = R\,\hat{c}_{G_0}, \\ \hat{c}_{G_0} &:= \big(\hat{c}_Z(\|\mathbf{s}_0 - \mathbf{s}_1\|_2), \ldots, \hat{c}_Z(\|\mathbf{s}_0 - \mathbf{s}_n\|_2)\big)^\top. \end{aligned} \tag{9}$$

The entry of $\hat{c}_0$ for estimate $(\mathbf{s}, j)$ is $\hat{c}_Z(\|\mathbf{s}_0 - \mathbf{s}\|_2)$, the same for every replicate at $\mathbf{s}$. Thus $\hat{\theta}_0$ never appears in $\hat{c}_0$, whereas $\hat{\theta}_1$ appears only in the entries with $\mathbf{s} = \mathbf{s}_0$.

Prediction error. Let $x_0 := X_{\text{ref}}(\mathbf{s}_0)$ and let the prediction target be the latent logit proportion

$$T_0 := x_0^\top \beta + Z(\mathbf{s}_0), \tag{10}$$

the value a study at $\mathbf{s}_0$ would measure without error. Write $\Sigma := \Sigma(\hat{\theta})$ and $\mathbf{e} := \mathbf{Y} - X\beta = RZ + \boldsymbol{\epsilon}$, so that $\operatorname{Var}(\mathbf{e}) = \Sigma$, $\operatorname{Cov}\big(\mathbf{e}, Z(\mathbf{s}_0)\big) = \hat{c}_0$ (which carries no $\hat{\theta}_0$, since $\boldsymbol{\epsilon} \perp Z$), and $\operatorname{Var}\big(Z(\mathbf{s}_0)\big) = \hat{c}_Z(0) = \hat{\theta}_1 + \sum_{i=2}^{s} \hat{\theta}_i$. Note that $\hat{\beta} - \beta = A\mathbf{e}$ with $A := (X^\top \Sigma^{-1} X)^{-1} X^\top \Sigma^{-1}$. Substituting $\mathbf{r} = \mathbf{e} - X(\hat{\beta} - \beta)$ into (9) and subtracting T_0 gives

$$\hat{Y}(\mathbf{s}_0) - T_0 = b_0^\top (\hat{\beta} - \beta) + \big[\hat{c}_0^\top \Sigma^{-1} \mathbf{e} - Z(\mathbf{s}_0)\big], \qquad b_0 := x_0 - X^\top \Sigma^{-1} \hat{c}_0, \tag{11}$$

the error from estimating β plus the kriging error. The prediction error variance is

$$\hat{\sigma}^2(\mathbf{s}_0) := \operatorname{Var}\big(\hat{Y}(\mathbf{s}_0) - T_0\big) = \underbrace{\hat{c}_Z(0) - \hat{c}_0^\top \Sigma^{-1} \hat{c}_0}_{\text{kriging variance of } Z(\mathbf{s}_0)} + \underbrace{b_0^\top \big(X^\top \Sigma^{-1} X\big)^{-1} b_0}_{\text{uncertainty in } \hat{\beta}}. \tag{12}$$